CLIP2RS: Leveraging Pretrained Vision-Language Model for Semantic Segmentation of Remote Sensing Images

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***Abstract*—Semantic segmentation of remote sensing (RS) images is a very challenging task due to the complicated characteristics such as diversity, complexity, and massiveness. Current research endeavors are predominantly centered on utilizing visual context information exclusively through meticulous architecture design, often overlooking significant semantic details. This oversight limits the efficacy in tackling the challenge of intraclass variations. In this article, we propose CLIP2RS, which is devised to leverage the pretrained vision-language model (VLM) for the semantic segmentation of RS images via the guidance of prior knowledge stored in the pretrained foundation model. Specifically, CLIP2RS utilizes a two-stage training strategy to overcome the domain gap challenge between natural images and RS images. A dual-granularity alignment framework that simultaneously aligns pixel-level local features and image-level global features is designed to alleviate the severe class sample imbalance problem. In addition, a novel prompting mechanism is effectively explored to fully harness the potential of CLIP textual descriptions. We conduct comprehensive experiments on the iSAID, Potsdam, and Vaihingen datasets, and the experimental results show that our proposed method achieves state-of-the-art performances, demonstrating its superiority.**

***Index Terms*—Remote sensing (RS), semantic segmentation, vision-language model (VLM).**

I. Introduction

REMOTE sensing (RS) image analysis plays an important role in geosciences, including urban planning [1], vegetation monitoring [2], military surveillance [3], disaster monitoring [4], and meteorological monitoring [5]. As one of the most critical technologies for automatic analysis of RS images, semantic segmentation refers to classifying each pixel of an image into a semantic category. High-resolution RS images hold complicated characteristics such as diversity, complexity, and massiveness, which make the semantic segmentation of RS images a very challenging task.

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In recent years, researchers have developed many accurate and efficient algorithms for RS semantic segmentation. Fully convolutional network (FCN) [6] was the first FCN proposed in the field of semantic segmentation, achieving end-to-end pixel-level semantic segmentation. Since then, numerous networks with improvements over FCN have been proposed [7], [8], [9], [10], [11]. These networks typically consist of two parts: an encoder and a decoder, which integrate high-level and low-level features through skip connections. However, due to the limitation of fixed local receptive fields, these methods do not fully utilize contextual information during the feature extraction process. Subsequently, the contextual exploration methods have gradually developed over the past few years, including multiscale context extraction [12], [13], [14], [15], [16], attention-based context exploration [17], [18], [19], and graph convolution network (GCN)-based context exploration [20], [21], [22]. In order to obtain more general and essential representations of RS features, ViTAE [23], [24] and RingMo [25] have employed masked image modeling (MIM) on large-scale satellite imagery datasets and large vision transformers (ViTs) to develop pretrained foundation models for complex RS scenes, which can be applied to downstream tasks such as RS semantic segmentation.

Due to RS images usually having a large angular distribution range from a bird's eye view, objects within the same category often exhibit diverse appearances in different scenes. As shown in Fig. 1(a) and (b), both images contain objects categorized as "ships," which exhibit substantial differences in shape, size, color, and position. The intraclass variation makes it challenging for a single visual feature to fully represent all instances within the same category. Existing RS semantic segmentation models mostly optimize the model with single visual information, ignoring the semantic consistency information in the text. In this article, we propose CLIP2RS to leverage both visual and linguistic knowledge restored in the pretrained vision-language model (VLM), i.e., CLIP [26], to benefit the visual representation learning for pixel-level semantic segmentation of RS Images. CLIP leverages contrastive learning on 400 million image-text pairs to align visual and textual features in a shared space, enabling it to simultaneously utilize both modalities and capture rich semantic context. This enables CLIP to acquire general and more transferable visual features, allowing it to efficiently handle intricate visual

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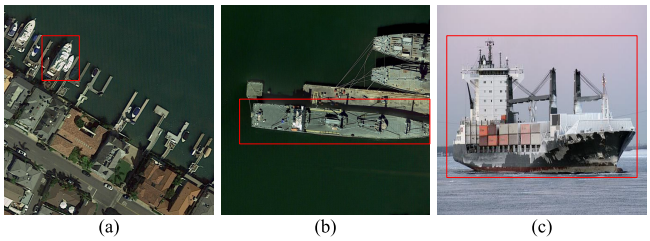


Fig. 1. View comparison of the same category in different scenes, where (a) and (b) are two RS images from the "ship" category, showing significant intraclass variance. (c) Natural image from the "ship" category, which has a distinct perspective and appearance difference compared to (a) and (b). Here, (a) and (b) are chosen from the iSAID dataset, while (c) is obtained from the ImageNet-1K dataset.

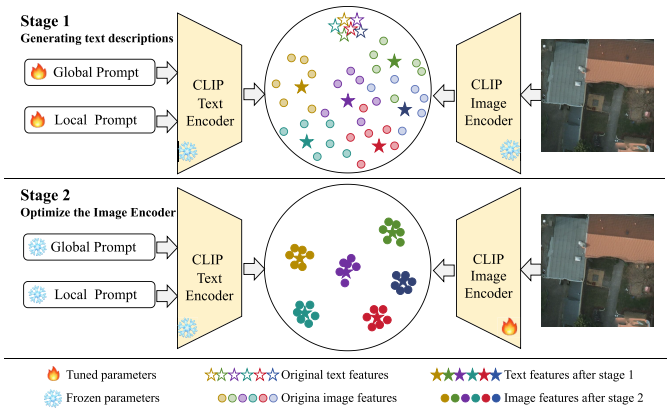


Fig. 2. Two-stage training strategy of our CLIP2RS. The first training stage aims to generate textual descriptions for various categories with learnable prompts, while the second training stage optimizes the image encoder under the guidance of these textual prompts.

elements and varied object appearances. Previous research [27], [28], [29] have demonstrated that multimodal VLM, i.e., CLIP have enormous potential in processing dense prediction tasks for natural images. However, the dataset used for pretraining CLIP primarily consists of natural images and text description pairs. The special sensors and wide shooting angles in the field of RS result in significant differences between RS images and natural images, as shown in Fig. 1(c) compared to Fig. 1(a) and (b). Directly applying the CLIP model to process RS images will face a great challenge of the domain gap, especially in dealing with the substantial intraclass variations and the complex environmental factors in RS data.

To overcome the above challenge, our CLIP2RS adopts a two-stage training strategy aimed at greatly augmenting the semantic descriptive ability of a single category label name via generating ample descriptive text prompts suitable for each category of RS images. Through the guidance of the abundant knowledge in the text encoder, the optimization of the visual encoder can be expected to learn a better visual representation. Specifically, a series of learnable context vectors with word embedding of the category label is combined as the textual prompt input to the text encoder. As can be seen in Fig. 2, in the first training stage, the parameters of the pretrained text encoder and image encoder are kept frozen, and only these learnable context vectors are optimized from scratch to adaptively generate textual prompt descriptions for each category. During the second training stage, the textual prompts and text encoder remain unchanged, while the image encoder is refined based on the local similarity map, which denotes the results of pixel-level visual-text matching, as illustrated in Fig. 3. Utilizing a segmentation head, the image encoder is enhanced to achieve high-resolution pixel-level alignment between visual and text features.

In addition, a high spatial resolution RS image typically encompasses various categories of objects, with significant variations in size. Large objects contribute to a significantly higher quantity of pixel samples compared to small objects, resulting in severe sample imbalance. Simply aligning pixel-level visual and textual features inevitably leads to decreased performance of models on small object categories. To address this issue, we further propose a dual-granularity alignment framework that aligns the two modalities at both fine-grained and coarse-grained levels. Besides the fine-grained pixel-level detailed alignment, the framework employs a multilabel classification loss to match visual and textual features at the coarse-grained image level. This dual-level alignment ensures a strong correspondence between pixel-level visual and textual features by reinforcing global consistency while preserving local details. By utilizing global information of the images to balance the gradient contributions of different classes for model optimization, the class imbalance problem, where the model may excessively focus on those classes with a large number of samples, can be significantly alleviated.

Furthermore, objects from different categories may exhibit both similar and unique characteristics. Traditional prompts contain learnable context vectors that are either entirely shared among all classes or entirely class-specific. We explore a novel design of prompts that can simultaneously capture general features and unique features for each category, fully unleashing the potential of CLIP textual description.

The main contributions are summarized as follows.

1) We introduce CLIP2RS, an approach utilizing VLM pretrained on natural images for the semantic segmentation of RS images. To address the domain gap issue between natural and RS images, we employ a two-stage training process.
2) We develop a dual-granularity alignment framework that concurrently aligns both local and global features, thereby addressing significant class sample imbalances. In addition, a new prompting mechanism is thoroughly investigated to maximize the effectiveness of CLIP textual descriptions.
3) The proposed method significantly outperforms the existing methods and achieves state-of-the-art performances on three widely adopted RS benchmark datasets, demonstrating the superiority of our method.

The rest of the article is organized as follows. Section II reviews the relevant works of our article. Section III elaborates on each component of the proposed method. In Section IV, we conduct thorough experiments on three widely used

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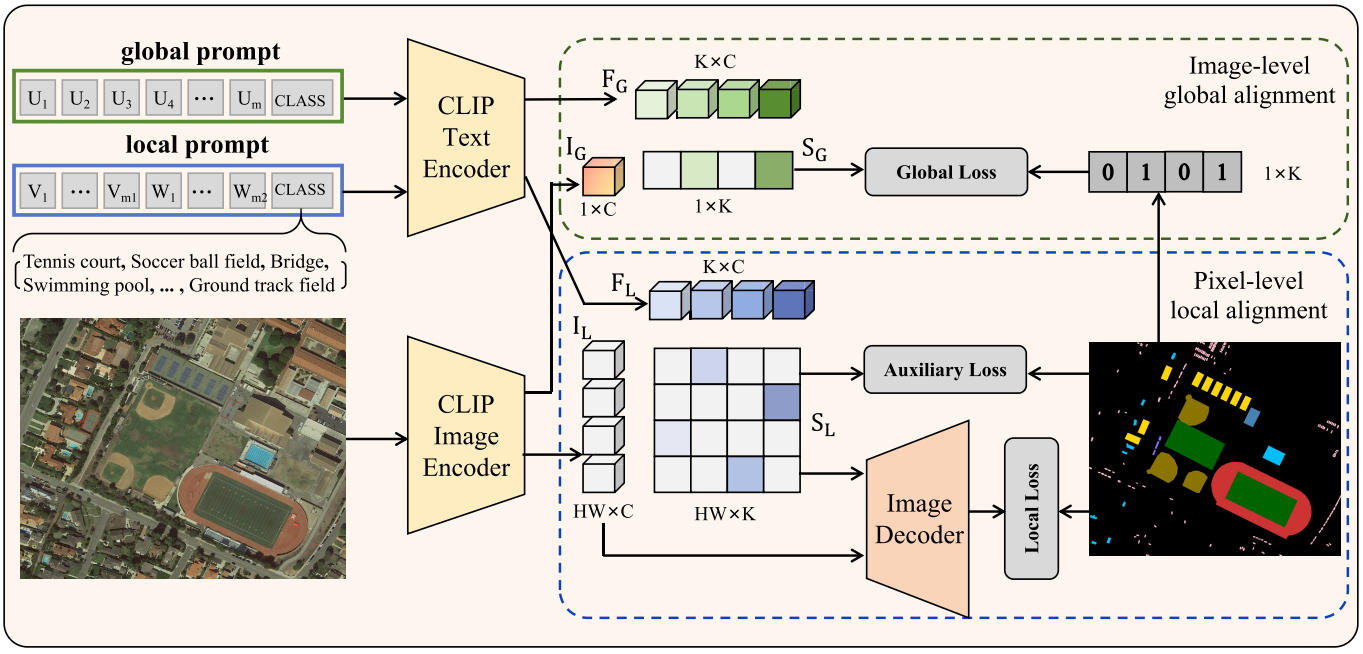


Fig. 3. Overall framework of our proposed dual-granularity alignment method. The dual-granularity alignment framework simultaneously aligns local features and global features, effectively alleviating the issue of severe class sample imbalance. In addition, a novel prompting mechanism to fully harness the potential of CLIP textual descriptions is well explored.

benchmark datasets to verify the effectiveness of our method. And we draw the conclusion of the article in Section V.

II. Related Work

A. Semantic Segmentation of RS Images

Semantic segmentation aims to assign a label to each pixel in an image, providing dense predictions for interpreting scenes. In recent years, substantial progress has been made in semantic segmentation techniques, starting with the introduction of FCNs [6]. FCNs [7] replace traditional fully connected layers in neural networks with convolutional layers, enabling end-to-end pixel-level predictions. However, due to the limited receptive field of these methods, they cannot fully utilize the context information.

Subsequently, many contextual exploration methods are proposed, including multiscale context extraction [12], [14], [30], attention-based context exploration [18], [19], [31], [32], and GCN-based context exploration [21], [22], [33]. Multiscale context extraction methods expand the receptive field by capturing contextual information at multiple scales, allowing the model to integrate both local and global features. HRFNet [15] introduces an entropy-based weighting and feature mining strategy to enhance feature extraction and fusion. DDFNet [16] integrates spatial and frequency domain features through dynamic decoupling and fusion and leverages geometric priors to enhance boundary precision. Attention-based context exploration methods explore long-distance dependencies by calculating the similarity between features or generating attention maps. ABCNet [19] designs a bilateral structure that adaptively captures rich spatial details in high-resolution images through spatial paths and simultaneously captures global contextual information through context paths. To further improve the context modeling, graph-based methods take feature maps as graph nodes and model their relationships through graph convolution operations. MSCGNet [21] extends the self-constructed graph (SCG) module to explicitly capture multiview global contextual representations in aerial images with rotation invariance.

In order to obtain more general and essential RS feature representations, some pretrained foundation models [23], [24], [25] for complex RS scenes have been proposed, which can be applied to downstream tasks such as RS semantic segmentation. ViTAE [23] proposes the first large-scale vision model suitable for RS tasks, presenting an innovative rotated, varied-size window attention mechanism as a replacement for the conventional full attention in transformers. RingMo [25] is the first generative self-supervised foundation model, which uses a large amount of RS data to obtain general feature representations and improves the accuracy of various RS tasks. Recently, SpectralX [34] introduced a parameter-efficient fine-tuning framework that adapts existing RS foundation models to multispectral and hyperspectral data. By employing a spectral-aware tokenizer and lightweight adapters, SpectralX effectively enhances domain generalization across diverse spectral modalities. Beyond foundation models, domain adaptation methods such as BiDA [35] propose a bi-directional transformer framework to jointly learn domain-invariant and domain-specific representations for cross-domain hyperspectral classification, providing complementary insights for robust segmentation under distribution shifts.

Different from the above methods that typically rely solely on visual information, our approach leverages both visual and linguistic knowledge from the pretrained VLM, i.e., CLIP [26]. This dual-modality integration enhances the learning of visual

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representations, leading to improved pixel-level semantic segmentation of RS images.

B. VLM for RS

Recently, VLMs have gained significant interest in the field of RS, primarily because language information offers a broader source of supervision for visual feature representations [26].

There are mainly two types of methods: contrastive VLMs-based methods and generative VLMs-based methods. Current contrastive VLMs-based methods usually build upon CLIP [26], which primarily consist of an image encoder and a text encoder. RemoteCLIP [36] learns robust visual features with rich semantics by utilizing aligned text embeddings, which are beneficial for downstream applications, such as image classification and object counting. RS-CLIP [37] leverages the pretrained CLIP model to learn semantic-aware visual representations through unified vision-language feature learning. Generative VLMs-based methods usually transform intricate visual features into a form that large language models (LLMs) can understand. SpectralGPT [38] introduces a novel 3-D generative pretrained transformer designed to efficiently process hyperspectral RS images. RSGPT [39] utilizes two LLMs, vicuna-7B and vicuna-13B, and incorporates an annotation-aware query transformer (Q-Former) to improve the alignment of visual and textual features. GeoChat [40] takes CLIP-ViT(L-14) as its visual backbone and enhances the transformer-based CLIP model by interpolating positional encoding.

In the context of RS semantic segmentation, SegCLIP [41] employs linguistic descriptions to enable CLIP to produce semantically distinct contextual details for RS imagery. MetaSegNet [42] extracts the key characteristic from the RS image metadata and transfers it into geographic text prompts via the generic ChatGPT. Then, a plug-and-play cross-modal attention fusion subnetwork is used to integrate heterogeneous image and text features.

Compared with existing methods that use VLMs for RS semantic segmentation, our method explores the use of CLIP, which is much cheaper to access as it is trained with natural images on the Internet. In addition, our proposed framework enables the model to focus on small sample categories through dual-granularity alignment at both global and local levels, fully leveraging CLIP's capabilities in RS dense prediction tasks.

C. Parameter-Efficient Fine-Tuning

To minimize training costs when adapting large pretrained models into downstream tasks, recent methods [43], [44], [45], [46], [47], [48], [49] learn continuous contexts to model prompts, called prompt tuning, to automate prompt engineering and explore optimal prompts.

Specifically, CoOp [43] demonstrates that the performance of CLIP is sensitive to prompts, and avoids manual prompt tuning by modeling context words with continuous vectors that are end-to-end learned from data. CoCoOp [44] further proposes to improve CoOp [43] by formulating the prompts in an image-conditional way to maintain better generalization to unseen classes. To avoid overfitting, ProGrad [45] leverages predictions from zero-shot CLIP to regularize gradients in the prompt learning process. ProDA [46] captures the distribution of diverse prompts to handle the varying visual representations and provides high-quality task-related content for facilitating recognition. TPT [47] suggests optimizing test-time prompts by promoting the consistency of augmented test images. DualCoOp [48] first adapts CLIP to multilabel image recognition with partially labeled data by learning pairs of positive and negative prompts for each class to ensure independent binary classification for each class.

Although existing prompt tuning approaches have achieved significant improvements in downstream tasks, they neglect to fully consider the commonalities and dependencies between different categories. In contrast, we propose a more robust prompt learning technique to compensate for this deficiency.

III. Methodology

In this section, we first introduce the training strategy of our CLIP2RS, then we elaborate on the overall framework, and finally, we demonstrate the new prompt design of our method.

A. Two-Stage Training Strategy

3CLIP is a powerful VLM that connects visual representations with their corresponding high-level linguistic descriptions. By pretraining on a dataset of 400 million (image, text) pairs collected from the internet, the model can perceive various high-level semantics from the text and learn transferable features. CLIP consists of two encoders, one is the image encoder, denoted as $\mathrm{Enc}_I(\cdot)$, the other is the text encoder, denoted as $\mathrm{Enc}_T(\cdot)$. The text encoder uses a transformer to encode textual information. The image encoder can be a CNN model, such as ResNet, or a ViT.

As shown in Fig. 2, during the first training stage, in order to retain the powerful feature extraction and generalization capabilities of the pretrained model, the parameters of the image encoder and text encoder from CLIP are kept fixed. In the visual end, input images pass through the image encoder to extract raw visual features. Simultaneously, on the text side, a series of learnable prompt vectors describing different categories is input to the text encoder to obtain corresponding text features. Then, under the supervision of pixel-wise labels, text features and visual features of the same category in the semantic space are aligned, aiming to make the text features of each category approach the central position of the corresponding visual features as closely as possible. With the back-propagation of gradients, the rich contextual information embedded in visual features is encoded into the parameters of the prompts, serving as textual descriptions for target categories in RS scenes. As shown in Fig. 2(a), pentagrams represent textual features of different categories, and triangles of the same color represent corresponding visual features. After the first stage of fine-tuning, distinctive textual features are generated among the discrete visual features.

During the second training stage, only the parameters of the image encoder are optimized. The text encoder and the prompts obtained in the first stage are frozen, jointly providing

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semantically relevant textual concepts for each category in RS images. This aims to supplement the supervised signal with language priors, offering constraints and guidance for rapidly fine-tuning the image encoder. Simultaneously, by combining the constraints of the objective function, it promotes vision-language cross-modal alignment in a shared semantic space. As shown in Fig. 2(b), after the second stage fine-tuning, visual features belonging to the same category are clustered around their corresponding textual features, while features from different categories are pushed further apart. This helps the model better distinguish between visual features of different categories.

This two-stage training strategy allows CLIP to effectively align text features and image features while retaining its original advantages. The textual descriptions specialized for each category in RS scenes are adaptively obtained through learnable context vectors, which then provide constraints for optimizing the image encoder, thereby enhancing the model's feature learning capability.

B. Dual-Granularity Alignment Framework

In order to align text features and visual features from the same category in a shared semantic space and alleviate the issue of sample imbalance, we design a dual-granularity alignment framework, as shown in Fig. 3. This framework handles local features and global features at two levels, respectively. Specifically, on one hand, pixel-level local alignment can learn fine-grained features of different categories to establish a connection between pixel-level visual features and text features. On the other hand, image-level global alignment helps balance the contribution of all categories to the gradients, avoiding excessive gradient bias toward categories with a large number of sample pixels during the model optimization process.

1) On the Text Side: Two sets of consecutive learnable context vectors are defined to combine with word vectors of class names as global prompts and local prompts, respectively. Global prompts are used to provide general descriptions of the target categories from the perspective of the entire image, while local prompts are used to specifically describe the detailed characteristics of the object categories. Formally, the two types of prompts can be represented as

$$\mathbf{P}_G = [U_1, U_2, U_3, \ldots, U_M, [\text{CLASS}]] \tag{1}$$

$$\mathbf{P}_L = [V_1, V_2, V_3, \ldots, V_M, [\text{CLASS}]] \tag{2}$$

where U_m and $V_m (m \in \{1, 2, \ldots, M\})$ are learnable context vectors, [CLASS] denotes the word vectors for the object classes, and $[\cdot, \cdot]$ denotes the concatenation operation. By forwarding these two sets of sequences to the text encoder $\text{Enc}_T(\cdot)$, we can obtain text features $\mathbf{F}_G$ and $\mathbf{F}_L$ corresponding to K categories (where K represents the total number of object categories)

$$\mathbf{F}_G = \text{Enc}_T(\mathbf{P}_G), \quad \mathbf{F}_L = \text{Enc}_T(\mathbf{P}_L), \ \mathbf{F}_G, \mathbf{F}_L \in \mathbb{R}^{K \times C}. \tag{3}$$

2) On the Image Side: An input image is divided into multiple equally sized patches, which are embedded into a latent space of dimension C along with position encoding to obtain the corresponding image embeddings $\mathbf{X}_L \in \mathbb{R}^{\text{HW} \times C}$. Together with an additional global vector ([CLS]) in the C-dimensional space $\mathbf{X}_G \in \mathbb{R}^{1 \times C}$, these image embeddings are used as input to the image encoder $\text{Enc}_I(\cdot)$. This process is represented as follows:

$$[\mathbf{I}_G, \mathbf{I}_L] = \text{Enc}_I([\mathbf{X}_G, \mathbf{X}_L]), \quad \mathbf{I}_G \in \mathbb{R}^{1 \times C}, \ \mathbf{I}_L \in \mathbb{R}^{\text{HW} \times C} \tag{4}$$

where $\mathbf{I}_G$ and $\mathbf{I}_L$ represent the image global feature and local features output by the image encoder.

3) Pixel-Level Local Alignment: The local features generated by the image encoder encode detailed contextual information of the objects in different scenes. Within a shared semantic space, similarity calculation is performed on the local features of the image and the corresponding local text features of the semantic classes to obtain a local similarity map $\mathbf{S}_L$

$$\mathbf{S}_L = \mathbf{I}_L \mathbf{F}_L^T, \quad \mathbf{S}_L \in \mathbb{R}^{\text{HW} \times K}. \tag{5}$$

The local similarity map describes the results of pixel-level visual-text matching. On one hand, it can be seen as low-resolution prediction results used to calculate auxiliary segmentation loss $\mathcal{L}_{\text{aux}}$. The $\mathcal{L}_{\text{aux}}$ quantifies the cross entropy between the local similarity map and the ground truth label. This encourages the feature map to faster restore its local structure, which is advantageous for dense prediction tasks. And on the other hand, it is connected with the last feature map output of the image decoder for calculating local segmentation loss $\mathcal{L}_{\text{local}}$. The objective functions both use cross-entropy loss, represented by the following equation:

$$\mathcal{L}_{\text{aux}} = \text{CrossEntropy}(\text{Softmax}(\mathbf{S}_L/\tau), \mathbf{Y}_L) \tag{6}$$

$$\mathcal{L}_{\text{local}} = \text{CrossEntropy}(\text{Softmax}(\text{Dec}_I([\mathbf{M}, \mathbf{S}_L])/\tau), \mathbf{Y}_L) \tag{7}$$

where $\tau = 0.07$ denotes the temperature coefficient following [50], $\mathbf{Y}_L \in \{1, 2, \ldots, K\}^{\text{HW}}$ denotes the ground truth pixel label, $\text{Dec}_I(\cdot)$ stands for the image decoder, and $\mathbf{M}$ is the last feature map output from the image decoder.

4) Image-Level Global Alignment: During the training process, the additional [CLS] vector dynamically integrates the information of all image patches into global semantic concepts through a mapping matrix generated by the cross-attention mechanism. The obtained global feature $\mathbf{I}_G$ can cover the semantics of the entire image, include information of all categories in the image, and match multiple text classes. Therefore, we treat it as a multilabel classification problem. To obtain the true situation of categories contained in the image, we use existing pixel-level annotation to obtain a K-dimensional binary vector, where 1 indicates the presence of the category in the current image, and 0 indicates its absence. Then, similarity calculation is performed between the global image feature and the global text features of all classes to obtain a K-dimensional similarity vector, as follows:

$$\mathbf{S}_G = \mathbf{I}_G \mathbf{F}_G^T, \quad \mathbf{S}_G \in \mathbb{R}^{1 \times K}. \tag{8}$$

Then, the Sigmoid operation is used to normalize $\mathbf{S}_G$, followed by using binary cross entropy loss as the objective function for the global branch, instead of using mutually exclusive loss like cross entropy, to ensure independent prediction for each class and avoid interference between different classes.

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The process for calculating the global segmentation loss $\mathcal{L}_{\text{global}}$ is as follows:

$$\mathcal{L}_{\text{global}} = \text{BinaryCrossEntropy}\left(\text{Sigmoid}\left(\mathbf{S}_G/\tau\right), \mathbf{Y}_G\right) \tag{9}$$

where $\mathbf{Y}_G \in \{0,1\}^K$ is the ground truth class label.

Under the supervision of $\mathcal{L}_{\text{global}}$, all existing categories in the same image are treated with equal probability. Through back-propagation, the gradient flow along the global feature $\mathbf{I}_G$ influences the optimization direction of the image encoder, thereby regulating the encoding of fine-grained visual features for each category.

The total loss of the whole framework is defined as

$$\mathcal{L}_{\text{task}} = \alpha\mathcal{L}_{\text{global}} + \beta\mathcal{L}_{\text{aux}} + \gamma\mathcal{L}_{\text{local}} \tag{10}$$

where α, β, and γ are coefficients to balance global loss, auxiliary loss, and local loss.

C. Double-Granularity Prompts

The textual prompts are used to extract semantic information for various categories in RS images, playing a crucial role in the process of model fine-tuning. In this article, we design double-granularity prompts to describe coarse-grained and fine-grained category information.

1) Coarse-Grained Global Prompt: The function of our global prompts is to identify multiple categories contained in the image from a coarse-grained perspective, so following CoOp, a unified context form is adopted, namely:

$$\mathbf{P}_G = [U_1, U_2, U_3, \ldots, U_M, [\text{CLASS}]] \tag{11}$$

where $[U_1, U_2, \ldots, U_M]$ denotes the context vectors shared among categories, *i.e.*, $[U_1^i, U_2^i, \ldots, U_M^i] = [U_1^t, U_2^t, \ldots, U_M^t]$ for $i \neq t$ and $i, t \in \{1, \ldots, K\}$. [CLASS] denotes the word vectors for the object classes, and the global prompt is defined by aggregating the class names of all objects within the image.

2) Fine-Grained Local Prompt: For fine-grained local prompts, different object categories in RS images exhibit both common features and unique differences. Relying simply on the [CLASS] token in the text feature space is insufficient to fully model the subtle differences between categories. Therefore, in order to fully capture the commonalities and differences among categories, we propose two sets of contexts, one set is a class-agnostic context shared among all categories, used to adaptively learn generic domain information, and the other set is a class-specific context used to model more subtle differences between categories, which are used to encode fine-grained features of that category, including typical shapes, textures, and contextual relationships

$$\mathbf{P}_L = [[V_1, \ldots, V_{M1}], [W_1, \ldots, W_{M2}], [\text{CLASS}]] \tag{12}$$

where $[V_1, \ldots, V_{M1}]$ denotes class-agnostic context vectors, i.e., $[V_1^i, \ldots, V_{M1}^i] = [V_1^t, \ldots, V_{M1}^t]$ for $i \neq t$ and $i, t \in \{1, \ldots, K\}$. $[W_1, \ldots, W_{M2}]$ denotes class-specific context vectors, i.e., $[W_1^i, \ldots, W_{M2}^i] \neq [W_1^t, \ldots, W_{M2}^t]$ for $i \neq t$ and $i, t \in \{1, \ldots, K\}$.

During the training process, we follow the two-stage training strategy proposed in Section III-A. First, the learnable context vectors in both the global prompt and local prompt are randomly initialized. In the first stage, the model parameters are frozen, and only the prompts are trained until the loss function converges. This process enables the prompts to capture text descriptions that reflect both the common features and the unique differences across different categories. In the second stage, the prompts trained in the first stage are loaded and kept fixed, and the text descriptions generated for specific RS categories are used to guide the optimization of the visual model.

IV. Experiments

A. Experimental Setup

1) Datasets: We evaluated the efficacy of our proposed method through experiments on three widely adopted open datasets: iSAID [51], ISPRS Potsdam, and ISPRS Vaihingen.

a) iSAID: The iSAID dataset is a semantic segmentation dataset and contains 2806 aerial images with resolutions ranging from 800 × 800 to 4000 × 13 000 pixels. This dataset is mainly collected from Google Earth, and some images are captured by the JL-1 satellite, while the others are captured by the GF-2 satellite. The whole dataset includes 655 451 instances for 15 object categories, and the remaining nonobject pixels are assigned as background. Therefore, this dataset has 16 categories in total. Among the 2806 high-resolution images, 1411, 458, and 957 images are separated as the training set, validation set, and test set, respectively. Since the annotations of the test set are unavailable, the validation set is used for testing in this article, which follows the evaluation protocols in all the comparison methods.

b) ISPRS Potsdam: The ISPRS Potsdam dataset published by ISPRS Commission WG II/4 consists of 38 high-resolution aerial images with a resolution of 0.5 m. The size of these images is 6000 × 6000 pixels, and there are 24 of them for training and 14 for testing. The dataset is annotated with six categories, including impervious surfaces, buildings, low vegetation, trees, cars, and clutter. For this dataset, images composed of near-infrared, red, and green spectral bands provided by the dataset are used for experiments.

c) ISPRS Vaihingen: The ISPRS Vaihingen dataset contains 33 very high-resolution orthophotography maps. The average size of the images in the dataset is 2494 × 2064 pixels, and there are 16 of them for training and 17 for testing. The dataset is annotated with six categories, including impervious surfaces, buildings, low vegetation, trees, cars, and clutter. The orthophoto images have three channels: the infrared channel, the red channel, and the green channel, each containing a wealth of spectral information.

The iSAID dataset provides a large-scale, multicategory dataset with varying resolutions, making it suitable for general-purpose segmentation tasks. In contrast, the ISPRS Potsdam and Vaihingen datasets focus on urban environments with detailed annotations and multispectral data, offering a unique combination of geographical types, spatial resolution, spectral information, and data volume. Together, these three datasets enable a more comprehensive assessment of model performance, covering a broad spectrum of segmentation challenges and ensuring a thorough evaluation of the model's capabilities.

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2) Evaluation Metrics: Considering evaluation criteria commonly used in segmentation, we report the intersection over union (IoU) for each foreground class (15 in total), and the mean IoU (mIoU) overall categories (16 in total, including background) for the iSAID dataset. As for the Potsdam dataset and the Vaihingen dataset, we report the $F1$ score per category, mean $F1$ score (m$F1$), overall accuracy (OA), and mIoU metrics. Note that the clutter category is ignored during model training and testing. The datasets are fully published, allowing models to be trained with training images and locally evaluated using test images.

3) Implementation Details: In our experiments, we consider three kinds of image backbones, including ResNet-50 [73], ResNet-101 [73], and ViT-B [74] as the image encoder, use the corresponding CLIP Transformer as the text encoder, and use the UperNet [57] as the image decoder. The total length of the learnable context vectors for both global and local prompts is set to 16 across all datasets. The analysis in Section IV-C indicates that when the lengths of the unified context and class-specific context are set to 8, respectively, the best results are achieved on most metrics across the datasets. In the training phase, the AdamW optimizer is utilized. The initial learning rate and weight decay are set to 0.0001 and 0.001, respectively. The polynomial decay rule is applied with a power of 0.9 and a minimal learning rate of 1×10^{-6}. The values of the loss weight coefficients α, β, and γ are set to 0.3, 0.3, and 0.4, respectively, which is proven to be the optimal hyperparameter combination in Section IV-C. To ensure the fairness of the results, all models are trained for 40 epochs with a batch size of 2 in iSAID, 200 epochs with a batchsize of 2 in the Potsdam dataset, and 600 epochs with a batchsize of 2 in the Vaihingen dataset. Because the size of the images varies greatly, we crop the images of iSAID into patches of 896×896 pixels with an overlap of 384 (stride is 512), and the images of Potsdam and Vaihingen are cropped into patches of 512×512 pixels with an overlap of 128 (stride is 384). For both datasets used, we employ some regular data augmentation strategies, including random rotation (90°, 180°, and 270°), random resizing (from 0.5 to 2.0), and the addition of random horizontal and vertical flipping. It is worth noting that unlabeled regions are not involved in either training or testing, given that these regions lack corresponding texts to form image-text pairs.

B. Comparison With the State-of-the-Arts Methods

1) Quantitative Evaluation:

a) iSAID: Table I reports the detailed comparison results of our CLIP2RS against other state-of-the-art methods on the iSAID dataset.

From Table I, we can get four conclusions.

1) Our proposed method outperforms all the comparison SOTA methods, achieving the highest mIoU of 68.0% on the ViT-B backbone.

2) We also achieve the highest scores for the vast majority of categories, especially on the categories with fewer samples. Although SegCLIP [41] demonstrates advantages in several categories, such as ground track field (GTF), Harbor, large vehicle (LV), and soccer ball field (SBF). We attribute this to its incorporation of all category names into the text descriptions, which effectively constrains the query space and mitigates misclassification for these specific classes. However, the absence of local semantic guidance limits its performance in other categories. In contrast, our method achieves improvements of 10.3%, 3.3%, 4.9%, 8.2%, and 2.2% in the categories of baseball diamond, basketball court, helicopter, swimming pool, and tennis court, respectively. These results highlight the effectiveness of the proposed dual-granularity alignment framework, which captures fine-grained details while simultaneously leveraging global textual semantics. This design helps alleviate class imbalance and leads to more consistent and superior performance across the majority of categories.

3) Compared with RS foundation models pretrained on large-scale RS datasets, our method also shows encouraging results on the iSAID dataset. Notably, our method outperforms UperNet-RingMo [25] and UperNet-ViTAEv2-S [24] by 0.8% and 3.7% in average performance mIoU, demonstrating promising performance using pretrained models from the natural image domain. This indicates that our introduction of CLIP to generate textual information for optimizing semantic segmentation is effective, and our proposed two-stage training strategy can alleviate the domain gap and transfer the powerful capabilities of the CLIP model to RS segmentation tasks. Not to mention that the CLIP model is much cheaper to access than those RS foundation models, which require a large collection of RS imagery.

4) We experiment with different backbones for the image encoder, and our method achieves superior performance when using the same backbones. The results show that our method consistently outperformed other methods using the same backbone, obtaining an improvement of 1. 0% mIoU over GRDL [64] on ResNet-50.

b) ISPRS Potsdam: Table II shows the segmentation results of our approach and other representative state-of-the-art methods on the Potsdam dataset. As in other works [82], [83], [84], clutter/background accuracy is not reported.

It can be obviously seen that as follows.

1) Our CLIP2RS far outperformed other models, achieving the highest m$F1$ of 93.80%, OA of 93.35% and mIoU of 88.32% with the ViT-B backbone. An improvement of 0.42% in m$F1$, 1.38% in OA, and 0.51% in mIoU compared with the second-best method HBCNet [83] can be observed. It is worth noting that our method outperforms the visual-language model MetaSegNet [42] by 0.60% in m$F1$, 1.25% in OA, and 0.82% in mIoU.

2) The $F1$ scores of CLIP2RS on small-scale objects, such as buildings, low vegetation, tree and cars are also the best. Especially on the car category, our method obtains an $F1$ score of 97.84%, surpassing the second-place by 0.84%, which indicates that our method has better results for small-scale objects.

3) When equipped with the same backbone, our method achieves much more competitive results than other SOTA methods. Our method improves the performance by 0.17% in $F1$ score compared with MFNet [75] on ResNet-50 backbone and 0.41% in $F1$ score compared with HMANet [60] on ResNet-101 backbone.

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TABLE I

RESULTS OF OUR MODEL WITH DIFFERENT BACKBONES AND SOTA METHODS ON THE VALIDATION SET OF THE iSAID DATASET. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING. "-" DENOTES NOT AVAILABLE

Methods	Backbone	mIoU(%)	IoU of each category(%)														
			BD	BC	Bridge	GTF	Harbor	HC	LV	Plane	RA	Ship	SV	SBF	ST	SP	TC
UNet [7]	-	39.2	36.5	22.9	7.5	5.5	45.6	0	49.9	74.7	46.5	49.0	35.6	9.7	0	38.0	78.6
FCN [6]	VGG-16	41.7	26.4	30.2	8.2	27.9	42.0	0	49.3	62.9	51.9	51.7	37.0	52.1	22.9	30.7	74.8
EMANet [52]	ResNet-50	55.4	66.2	56.0	42.1	18.8	47.5	33.4	58.2	78.5	46.9	63.1	41.0	46.4	68.4	38.9	82.7
DANet [53]	ResNet-50	57.5	71.4	50.9	28.6	52.5	48.8	30.4	57.5	80.9	40.6	60.2	42.1	63.3	63.0	46.1	84.7
CCNet [54]	ResNet-50	58.3	68.9	57.1	34.0	56.8	45.9	31.6	57.6	75.8	57.2	61.4	38.3	75.0	65.7	36.5	82.9
EncNet [13]	ResNet-50	58.9	70.0	55.2	36.8	46.3	48.0	34.8	57.2	76.1	59.8	59.7	38.7	69.8	64.9	42.4	84.2
Semantic FPN [55]	ResNet-50	59.3	71.8	57.8	34.0	51.6	51.3	0	59.2	80.8	68.7	63.7	45.1	73.6	59.5	46.4	86.6
RefineNet [9]	ResNet-50	60.2	72.3	61.1	32.6	52.8	51.1	23.0	58.2	79.9	65.6	63.8	42.4	74.4	58.6	43.4	85.3
PSPNet [8]	ResNet-50	60.3	75.7	61.1	32.5	60.2	54.3	10.9	58.0	79.5	68.6	65.2	43.0	71.9	52.1	46.8	85.6
DeeplabV3+ [56]	ResNet-50	60.8	72.8	56.5	32.2	58.9	52.6	31.4	59.1	79.8	67.7	63.9	42.9	72.9	52.5	46.1	84.9
UperNet [57]	ResNet-50	61.9	68.1	57.3	39.2	52.5	52.2	34.3	61.2	83.8	62.1	65.9	48.8	76.8	73.9	44.5	70.7
AlignSeg [58]	ResNet-50	62.1	76.2	62.1	28.7	52.0	54.8	31.2	60.7	82.9	56.2	67.4	50.3	71.2	68.9	45.7	86.2
RANet [59]	ResNet-50	62.1	72.5	53.2	45.3	47.1	55.6	38.1	60.1	83.1	70.5	67.1	49.3	58.8	61.3	41.8	85.1
HMANet [60]	ResNet-50	62.6	74.7	60.5	29.0	54.6	51.9	32.6	59.7	83.8	62.9	65.4	50.3	70.2	70.9	51.4	88.7
GSCNN [61]	ResNet-50	63.4	72.6	56.1	40.7	58.4	54.4	33.8	63.8	63.8	58.5	65.9	51.1	72.5	71.2	48.8	85.5
FarSeg [62]	ResNet-50	63.7	77.7	62.1	36.7	56.7	53.9	35.8	60.6	82.0	71.4	65.4	46.3	72.5	61.8	51.2	86.4
FactSeg [63]	ResNet-50	64.8	78.4	64.9	36.3	54.6	55.7	42.7	62.7	84.1	69.4	68.3	49.5	73.6	56.8	51.5	88.9
GRDL [64]	ResNet-50	65.1	78.5	63.5	38.1	56.9	52.8	38.6	60.5	84.2	68.7	65.5	50.7	73.7	69.2	52.4	88.7
MP-Former [65]	ResNet-50	63.7	67.4	60.6	44.1	58.6	57.0	22.5	65.1	84.9	75.8	66.7	53.5	62.4	67.9	48.5	85.3
DenseCLIP [66]	ResNet-50	59.2	61.2	61.5	36.8	50.7	46.4	23.0	53.0	73.2	52.0	64.1	47.9	76.1	55.1	62.2	85.6
SegCLIP [41]	ResNet-50	65.4	66.9	58.6	44.8	59.2	59.2	32.2	65.2	85.4	75.6	70.5	53.5	74.3	68.1	46.8	86.9
CMTFNet [67]	ResNet-101	63.6	77.0	61.4	40.2	62.7	55.5	16.9	60.0	80.5	72.7	65.0	43.7	76.7	71.7	47.4	87.6
RSSFormer [68]	RSS-B	65.6	**79.7**	64.4	44.8	61.1	55.6	35.1	60.7	73.6	**78.3**	65.9	41.3	78.6	73.6	50.2	86.9
DenseASPP [69]	DenseNet-121	56.8	67.5	56.6	29.6	52.3	51.1	0	57.1	78.1	64.8	61.1	38.4	74.1	50.0	43.3	86.1
DenseUNet [70]	DenseNet-121	58.7	76.1	57.7	33.9	49.5	54.6	0	54.7	82.2	65.9	66.1	46.2	71.9	50.4	45.1	86.2
HRNet [71]	HRNetV2-W48	61.5	74.0	59.4	33.8	61.5	52.2	14.9	62.1	81.7	52.9	65.9	46.9	75.6	68.9	44.2	86.9
OCR [72]	HRNetV2-W48	62.6	73.6	63.4	33.1	47.7	53.3	30.4	61.4	83.3	59.5	67.8	49.6	72.8	70.7	48.4	87.9
UperNet-ViTAEv2-S [24]	ViTAEv2-S	64.3	72.2	57.4	44.0	63.0	54.7	35.4	62.5	85.2	62.2	71.3	51.6	77.7	74.3	47.0	70.4
UperNet-RingMo [25]	Swin-B	67.2	77.0	62.5	43.2	63.0	58.9	40.1	63.9	85.7	67.3	73.5	51.2	78.5	73.0	48.9	89.1
SegCLIP [41]	Swin-B	67.8	69.0	62.5	45.6	**65.5**	**62.8**	38.6	**68.2**	**86.6**	74.9	73.5	**54.3**	**79.1**	74.2	45.0	87.7
Ours (CLIP2RS)	ResNet-50	66.1	76.0	60.1	42.2	61.9	58.4	42.1	64.3	85.6	70.7	70.7	52.0	74.1	63.0	50.8	87.1
Ours (CLIP2RS)	ResNet-101	66.6	75.9	64.9	37.9	58.9	59.3	42.7	65.3	86.3	71.0	**73.6**	53.5	74.3	67.7	50.5	87.5
Ours (CLIP2RS)	ViT-B	**68.0**	79.3	**65.8**	**45.7**	57.8	55.7	**43.5**	65.7	85.6	72.5	72.1	54.1	74.6	**74.7**	**53.2**	**89.9**

[1] The 15 foreground categories of iSAID dataset: baseball diamond (BD), basketball court (BC), Bridge, ground track field (GTF), Harbor, helicopter (HC), large vehicle (LV), Plane, roundabout (RA), Ship, small vehicle (SV), soccer ball field (SBF), storage tank (ST), swimming pool (SP) and tennis court(TC).

c) ISPRS Vaihingen: Table III reported the segmentation results of our CLIP2RS and other SOTA methods on the Vaihingen dataset. As with the Potsdam dataset, clutter/background accuracy is not reported. It can be concluded that as follows.

1) The results indicate that our proposed method demonstrates significant improvements in terms of both m$F1$, OA, and mIoU metrics as compared to all other methods. There is a 0.45% improvement in m$F1$, 0.38% improvement in OA, and 0.31% improvement in mIoU compared with the second-best method CTMFNet [82].

2) And we obtain the highest $F1$ scores of 93.82%, 96.99%, 86.25%, and 91.78% for most categories of the impervious surface, building, low vegetation, and car, respectively.

3) Moreover, for the same backbone, our method achieves much more competitive results than other SOTA methods on both ResNet-50 and ResNet-101 backbones, showing the consistent superiority of our method.

2) Qualitative Evaluation:

a) iSAID: Fig. 4 illustrates a visual comparison between the results produced by the baseline method and our approach on the iSAID dataset. Throughout all subfigures, the regions in the red boxes represent the challenging segmentation areas. As shown in the first row, for small-scale object storage tanks with very few samples, CLIP2RS is able to identify and segment them correctly, while other methods mistakenly predict them as other categories. This indicates that we can effectively balance the gradient contribution of different categories to model optimization by utilizing the global information of images, thereby boosting the performance of the model on a few sample categories. Furthermore, our method successfully addresses the intraclass variance. As can be seen from the second and third rows, the scale and appearance of the ship category vary greatly in different scenes. Our CLIP2RS is still able to demonstrate better segmentation results compared to other methods, proving that the semantic information contained in the text can indeed promote the learning of visual representations.

b) ISPRS Potsdam: Fig. 5 shows the visualization results of our method and several representative compared methods on the Potsdam dataset. Intuitively, the trees and low vegetation are staggered and have similar colors, making it difficult to distinguish, as shown in the red boxes in the second row. For these challenging categories, our method is able to identify

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TABLE II

RESULTS OF OUR MODEL WITH DIFFERENT BACKBONES AND SOTA METHODS ON THE VALIDATION SET OF THE POTSDAM DATASET. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING. "-" DENOTES NOT AVAILABLE

Methods	Backbone	mF1(%)	OA(%)	mIoU(%)	F1 score per category(%)				
					Imper.surf.	Building	Low veg.	Tree	Car
AlignSeg [58]	ResNet-101	76.39	75.74	62.24	79.82	88.53	68.42	71.52	73.66
CCNet [54]	ResNet-101	83.85	82.45	72.13	84.74	90.09	79.78	82.03	82.61
EMANet [52]	ResNet-101	79.64	78.20	66.34	80.15	87.40	75.60	78.25	76.80
FarSeg [62]	ResNet-101	83.62	82.48	72.0	86.12	90.79	78.92	77.88	84.42
UperNet [57]	ResNet-101	77.61	76.80	63.48	81.30	86.59	68.21	78.69	73.26
RSSFormer [68]	ResNet-101	87.89	86.65	78.39	89.22	94.22	82.72	84.01	89.27
UNet [7]	-	92.28	90.63	85.94	93.03	96.79	87.01	88.02	96.53
FCN [6]	VGG-16	91.44	89.63	84.48	92.07	95.66	86.27	87.44	95.75
MFNet [75]	ResNet-50	93.25	91.96	87.57	94.25	97.52	88.42	89.43	96.62
MANet [76]	ResNet-101	92.90	91.32	86.95	93.40	96.96	88.32	89.36	96.48
DANet [53]	ResNet-101	91.48	89.98	84.57	91.96	96.35	86.20	87.21	95.92
DeepLabV3+ [56]	ResNet-101	92.12	90.88	84.32	92.95	95.88	87.62	88.15	96.02
PSPNet [8]	ResNet-101	92.40	91.08	84.88	93.36	96.97	87.75	88.50	95.42
CCANet [77]	ResNet-101	92.41	91.47	85.65	93.58	96.77	86.87	88.59	96.24
HMANet [60]	ResNet-101	93.20	92.21	87.28	93.85	97.56	88.65	89.12	96.84
ABCNet [19]	ResNet-18	87.67	-	78.51	90.05	94.53	81.87	79.61	92.30
UNetformer [78]	ResNet-18	91.19	-	84.15	92.40	96.54	85.34	86.07	95.59
TransUNet [79]	R50-ViT-B	90.69	-	83.31	92.06	95.95	85.05	85.28	95.13
RS3Mamba [80]	R18-MambaT	91.36	-	84.45	92.37	96.87	85.42	86.23	95.90
SSNet [81]	ViT-B	91.12	-	84.02	91.98	96.58	85.32	85.93	95.78
CMTFNet [67]	ResNet-50	91.73	-	85.02	92.46	96.81	86.30	86.95	96.11
CTMFNet [82]	HRNet&transformer	92.84	91.38	86.85	93.22	97.12	87.87	89.36	96.61
HBCNet [83]	HRNetV2-W48	93.38	91.97	87.81	94.29	97.54	88.49	89.58	97.00
DC-Swin [84]	Swin-S	93.25	92.00	87.56	94.19	97.57	88.57	89.62	96.31
FTUNetformer [78]	Swin-B	86.35	-	76.54	88.64	93.25	80.74	76.81	92.29
MetaSegNet [42]	Swin-B	93.20	92.10	87.50	**94.60**	97.30	88.10	89.70	96.40
Ours (CLIP2RS)	ResNet-50	93.42	92.86	87.64	93.69	97.41	89.17	89.47	97.35
Ours (CLIP2RS)	ResNet-101	93.61	93.02	87.95	93.84	97.68	89.46	89.51	97.56
Ours (CLIP2RS)	ViT-B	**93.80**	**93.35**	**88.32**	94.10	**97.83**	**89.52**	**89.73**	**97.84**

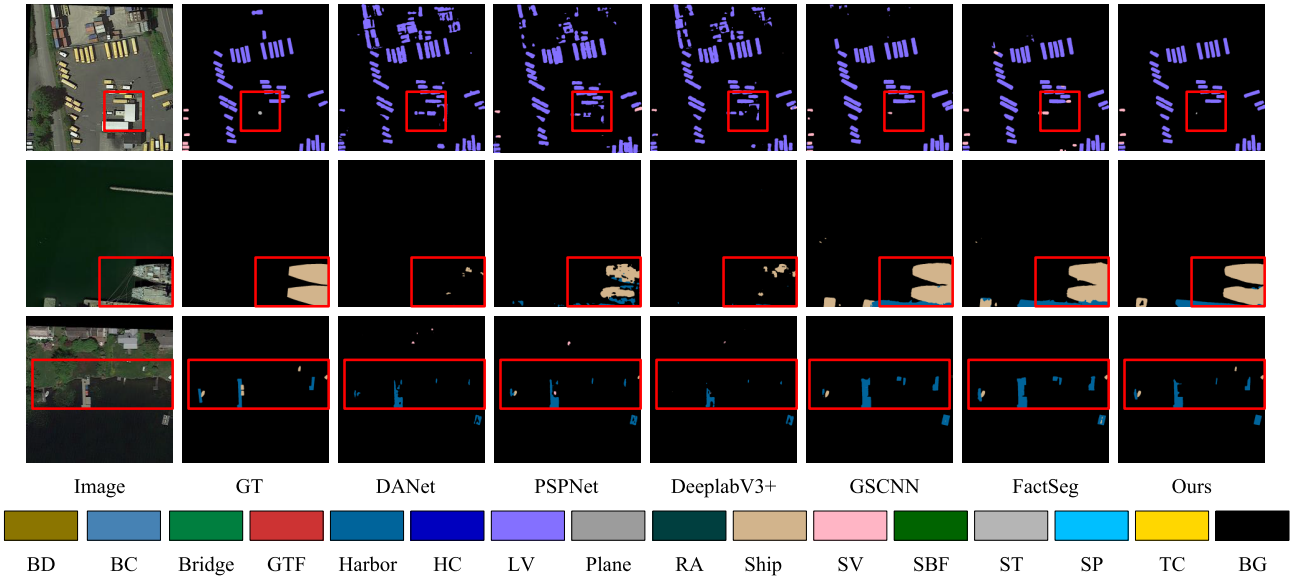


Fig. 4. Qualitative comparison between our method (CLIP2RS) and other methods on the iSAID dataset. The region in the red box represents the challenging segmentation area.

them as accurately as possible, achieving relatively better prediction results than currently popular methods. In addition, our approach excelled in precisely segmenting entire objects, notably evident in the clear delineation and complete contours

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TABLE III

RESULTS OF OUR MODEL WITH DIFFERENT BACKBONES AND SOTA METHODS ON THE VALIDATION SET OF THE VAIHINGEN DATASET. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING. "-" DENOTES NOT AVAILABLE

Methods	Backbone	mF1(%)	OA(%)	mIoU(%)	F1 score per category(%)				
					Imper.surf.	Building	Low veg.	Tree	Car
AlignSeg [58]	ResNet-101	82.82	85.69	70.68	87.46	92.05	78.51	84.91	71.18
CCNet [54]	ResNet-101	80.68	81.50	67.50	85.74	90.66	76.29	83.80	66.93
EMANet [52]	ResNet-101	72.16	75.88	56.45	83.08	85.84	67.84	64.05	59.99
FarSeg [62]	ResNet-101	74.32	75.00	59.26	82.83	86.18	75.17	82.87	45.55
UperNet [57]	ResNet-101	70.70	63.97	54.68	78.39	81.35	64.86	80.46	48.46
RSSFormer [68]	ResNet-101	84.09	85.95	72.54	88.45	92.58	77.72	84.74	76.97
UNet [7]	-	89.39	90.41	81.08	92.66	95.25	83.52	89.36	86.16
FCN [6]	VGG-16	88.47	89.14	79.58	91.27	94.16	81.81	88.52	86.59
MFNet [75]	ResNet-50	90.88	91.67	83.50	93.43	96.35	85.85	90.50	88.31
MANet [76]	ResNet-101	90.41	90.96	82.71	93.02	95.47	84.64	89.98	88.95
DANet [53]	ResNet-101	89.19	90.44	81.32	91.63	95.02	83.25	88.87	87.16
DeepLabV3+ [56]	ResNet-101	89.57	90.56	81.47	92.38	95.17	84.29	89.52	86.47
PSPNet [8]	ResNet-101	90.26	90.85	82.58	92.79	95.46	84.51	89.94	88.61
CCANet [77]	ResNet-101	90.58	91.11	82.76	93.29	95.53	85.06	90.34	88.70
HMANet [60]	ResNet-101	90.96	91.44	83.49	93.50	95.86	85.41	90.40	89.63
ABCNet [19]	ResNet-18	85.34	-	75.20	89.78	94.30	78.49	90.08	74.05
UNetformer [78]	ResNet-18	89.85	-	81.97	92.33	96.25	80.47	90.85	89.35
TransUNet [79]	R50-ViT-B	87.46	-	78.16	90.77	94.32	79.02	90.53	82.66
RS3Mamba [80]	R18-MambaT	90.34	-	82.78	92.83	96.82	80.84	**91.10**	90.09
SSNet [81]	ViT-B	89.87	-	82.04	92.06	96.79	80.53	91.07	88.10
CMTFNet [67]	ResNet-50	89.91	-	82.12	92.53	96.95	79.98	90.22	89.87
CTMFNet [82]	HRNet&transformer	91.37	91.60	84.34	93.79	96.12	85.02	90.47	91.47
HBCNet [83]	HRNetV2-W48	91.32	91.72	84.21	93.60	96.13	85.95	90.53	90.40
DC-Swin [84]	Swin-S	90.71	91.63	83.22	93.60	96.18	85.75	90.36	87.64
FTUNetformer [78]	Swin-B	90.24	-	82.62	93.41	96.92	81.53	90.91	88.46
Ours (CLIP2RS)	ResNet-50	90.99	91.72	83.86	92.87	96.75	85.51	89.36	90.48
Ours (CLIP2RS)	ResNet-101	91.20	91.85	84.12	93.05	96.94	85.65	89.71	90.66
Ours (CLIP2RS)	ViT-B	**91.82**	**91.98**	**84.65**	**93.82**	**96.99**	**86.25**	90.28	**91.78**

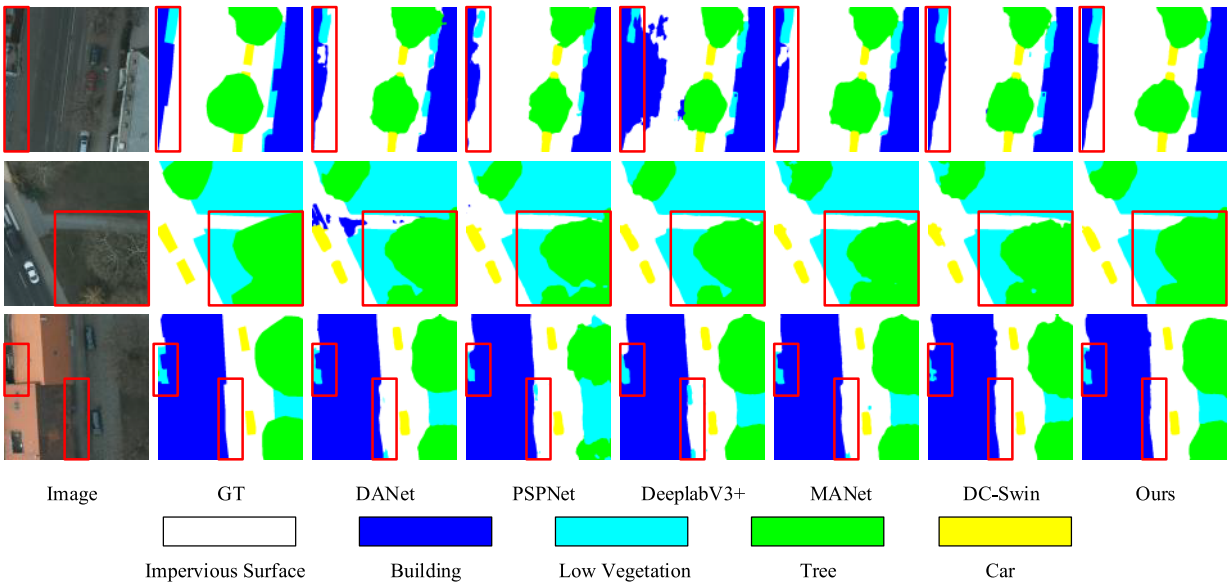


Fig. 5. Qualitative comparison between our method (CLIP2RS) and other methods on the Potsdam dataset. The region in the red box represents the challenging segmentation area.

of the building shown in the red rectangular boxes in the first and third rows. These observations closely align with the improvements indicated by the $F1$ score per category as presented in Table II.

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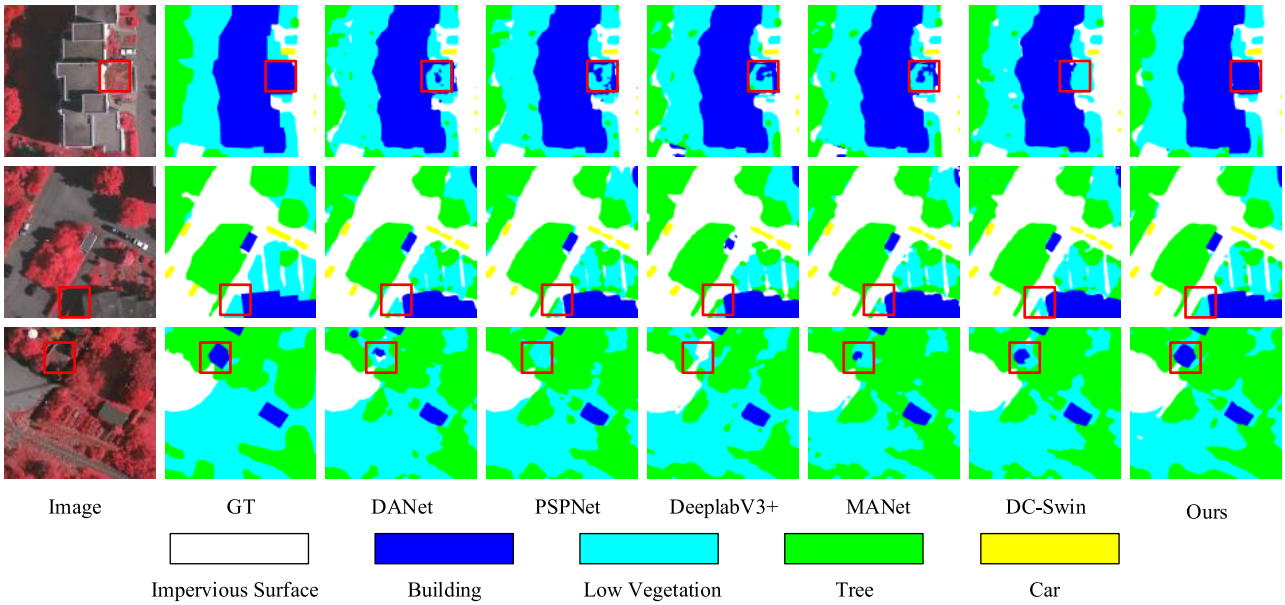


Fig. 6. Qualitative comparison between our method (CLIP2RS) and other methods on the Vaihingen dataset. The region in the red box represents the challenging segmentation area.

c) ISPRS Vaihingen: Fig. 6 shows the visualization results of our method and several representative methods on the Vaihingen dataset. For the small area of low vegetation highlighted in the red box in the second row, other methods are difficult to distinguish between low vegetation and trees, resulting in missed or false alarms. In contrast, our method can handle this situation perfectly. For buildings with different appearances, as shown in the red boxes in the first and third rows, it is clear that our model is able to accurately and completely segment the objects. However, other methods tend to fail when faced with such challenging cases. Above improvements primarily stem from our dual-granularity alignment framework and text prompt information, which enhances the semantic correlation between pixels of the same category.

3) Summary and Discussion: Overall, the proposed method outperforms existing single-visual modality approaches in RS semantic segmentation, including large models specifically designed for RS tasks, such as UperNet-RingMo [25] and UperNet-ViTAEv2-S [24]. CLIP2RS sets new records across multiple datasets, demonstrating the effectiveness of incorporating textual guidance. We attribute this improvement to the semantic consistency of textual information, which enables flexible representation of diverse target features and prior knowledge, thereby enhancing the model's adaptability to the variability of RS targets and complex backgrounds.

Moreover, our method surpasses recent VLMs for RS segmentation, such as SegCLIP [41] and MetaSegNet [42]. This achievement is enabled by the two-stage training strategy we introduce, which effectively transfers CLIP's image-text alignment capability from natural images to RS images. In addition, the proposed dual-granularity alignment framework and dual-granularity prompt mechanism significantly enhance CLIP's performance in RS semantic segmentation tasks, significantly improving segmentation accuracy for small targets.

C. Further Analysis

In this section, we evaluate the effectiveness of our proposed components, which include a two-stage training strategy, a dual-granularity alignment framework, and the novel design of textual prompts. Ablation experiments are performed on three benchmark datasets, and the ViT-B is adopted as the backbone network.

1) Necessity of Two-Stage Training Strategy: To verify the necessity of our two-stage training strategy, we experimented with the one-stage training, in which we optimized the image encoder and text prompts at the same time. The results on three datasets are shown in Tables IV–VI. It can be seen that the two-stage training strategy significantly outperforms the single-stage training strategy on all datasets. For example, on the iSAID dataset, under the dual granularity framework and single granularity framework settings, the two-stage training improves mIoU of 3.03% and 3.24% compared to the one-stage training, respectively. The reason for this is probably because in the early stage of simultaneous training, learnable text tokens are not yet able to describe the images well, which may hamper the optimization of the image encoder.

2) Effectiveness of Dual-Granularity Alignment Framework: In order to emphasize the contribution of the global branch, we compare the dual-granularity alignment framework with its single-granularity variant, which aligns pixel-level visual-text features only from the fine-grained level. The results on three datasets are reported in Tables IV–VI. And the value below the class name represents the proportion of pixels that each class occupies in the foreground. The results underscore that under the same training strategy, the dual-granularity alignment framework exhibits superior performance compared

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TABLE IV

COMPARISON OF DIFFERENT TRAINING STRATEGIES AND OVERALL FRAMEWORKS ON THE iSAID DATASET. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING

Strategy	Framework	mIoU(%)	IoU of each category(%)														
			BD 1.9%	BC 2.7%	Bridge 1.4%	GTF 7.6%	Harbor 9.6%	HC 0.3%	LV 11.6%	Plane 9.7%	RA 4.6%	Ship 12.4%	SV 12.3%	SBF 15.9%	ST 1.7%	SP 1.1%	TC 7.2%
one-stage	single-granularity	64.8	72.5	55.2	41.7	58.4	57.2	40.0	65.3	85.2	69.1	71.2	53.9	73.1	61.0	47.9	87.1
one-stage	dual-granularity	66.0	76.1	60.4	42.4	56.7	57.5	42.2	65.6	85.7	70.9	70.7	52.6	74.3	65.0	50.1	88.2
two-stage	single-granularity	66.9	77.6	62.9	43.8	**59.4**	**57.8**	40.4	64.7	**86.9**	71.5	70.6	53.3	**75.8**	71.1	48.1	88.8
two-stage	dual-granularity	**68.0**	**79.3**	**65.8**	**45.7**	57.8	55.7	**43.5**	**65.7**	85.6	**72.5**	**72.1**	**54.1**	74.6	**74.7**	**53.2**	**89.9**

[1] The 15 foreground categories of iSAID dataset: baseball diamond (BD), basketball court (BC), Bridge, ground track field (GTF), Harbor, helicopter (HC), large vehicle (LV), Plane, roundabout (RA), Ship, small vehicle (SV), soccer ball field (SBF), storage tank (ST), swimming pool (SP) and tennis court(TC).

TABLE V

COMPARISON OF DIFFERENT TRAINING STRATEGIES AND OVERALL FRAMEWORKS ON THE POTSDAM DATASET. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING

Strategy	Framework	mF1(%)	OA(%)	mIoU(%)	F1 score per category(%)				
					Imper.surf. 33.0%	Building 26.6%	Low veg. 23.6%	Tree 15.1%	Car 1.7%
one-stage	single-granularity	92.78	92.75	86.89	93.34	97.12	88.58	88.78	96.06
one-stage	dual-granularity	93.23	92.88	87.83	93.85	97.54	88.85	88.96	96.94
two-stage	single-granularity	93.25	93.02	87.94	93.66	**97.93**	88.82	89.08	96.75
two-stage	dual-granularity	**93.80**	**93.35**	**88.32**	**94.10**	97.83	**89.52**	**89.73**	**97.84**

TABLE VI

COMPARISON OF DIFFERENT TRAINING STRATEGIES AND OVERALL FRAMEWORKS ON THE VAIHINGEN DATASET. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING

Strategy	Framework	mF1(%)	OA(%)	mIoU(%)	F1 score per category(%)				
					Imper.surf. 28.1%	Building 25.9%	Low veg. 20.9%	Tree 23.9%	Car 1.2%
one-stage	single-granularity	90.44	91.17	83.25	93.33	96.28	84.16	89.14	89.29
one-stage	dual-granularity	91.06	91.56	83.49	93.70	96.62	84.55	89.90	90.55
two-stage	single-granularity	91.39	91.59	84.30	93.73	96.69	**86.57**	90.15	89.83
two-stage	dual-granularity	**91.82**	**91.98**	**84.65**	**93.82**	**96.99**	86.25	**90.28**	**91.78**

to the single-granularity alignment framework on all datasets.

Specifically, as can be seen from Table IV that for the iSAID dataset, when using the two-stage training strategy, the dual-granularity alignment framework improves IoU by 2.19%, 4.61%, 4.34%, 7.67%, 5.06%, and 10.60% in the categories of baseball diamond, basketball court, bridge, helicopter, storage tank and swimming pool, respectively. And when using the one-stage training strategy, the dual-granularity alignment framework improves IoU by 4.97%, 9.42%, 1.68%, 5.50%, 6.56%, and 4.59% in the categories of baseball diamond, basketball court, bridge, helicopter, storage tank, and swimming pool, respectively. The results indicate that our method has a significant improvement for the few-sample categories. In addition, the results of Tables V and VI underscore that for the few-sample category of "car" in the Potsdam and Vaihingen datasets, the dual-granularity alignment framework brings 1.13% and 2.17% gains in $F1$ score under the two-stage training strategy on Potsdam and Vaihingen datasets, respectively.

The above observations prove that our method significantly improves the segmentation performance of objects with few-sample categories, while not excessively reducing the model's performance on other categories, thus boosting the overall performance. These improvements are primarily attributed to the global branch, which balances the gradient contributions of all categories by aligning the [CLS] token of the image encoder and the global text prompt.

3) Exploration of Different Designs of Prompts: In order to demonstrate the effectiveness of our proposed novel prompt design and explore which combination of double-granularity prompts is better, we attempt 12 prompt combinations on the Potsdam and Vaihingen datasets, respectively. The results are shown in Table VII.

We attempt the following four types of prompt designs: 1) UC: Unified context where context vectors are shared

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TABLE VII

COMPARISON OF DIFFERENT DESIGNS OF DOUBLE-GRANULARITY PROMPTS ON THE POTSDAM AND VAIHINGEN DATASETS. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING

Global	Local	Potsdam			Vaihingen		
		mF1(%)	OA(%)	mIoU(%)	mF1(%)	OA(%)	mIoU(%)
UC	UC	86.99	84.82	79.66	83.56	84.11	77.01
	CSC	93.39	93.17	88.10	90.82	91.44	83.52
	HC_CSC	93.44	93.30	88.16	91.59	91.95	84.33
	UC_CSC	**93.80**	**93.35**	**88.32**	**91.82**	**91.98**	**84.65**
CSC	UC	85.26	84.63	77.17	82.72	83.39	75.98
	CSC	93.19	93.02	87.39	89.68	90.99	82.39
	HC_CSC	93.22	93.08	87.67	89.76	91.11	82.67
	UC_CSC	93.33	93.13	88.08	90.12	91.17	82.82
HC_CSC	UC	86.57	84.75	79.23	82.83	83.64	76.24
	CSC	93.28	93.07	87.82	90.20	91.32	83.08
	HC_CSC	93.34	93.13	88.10	90.62	91.46	83.13
	UC_CSC	93.43	93.21	88.13	90.74	91.56	83.35
UC_CSC	UC	86.33	84.78	78.39	83.15	84.02	76.63
	CSC	93.25	93.01	87.92	90.18	91.25	82.92
	HC_CSC	93.33	93.10	88.05	90.35	91.37	83.10
	UC_CSC	93.53	93.25	88.17	91.25	91.59	84.17

among all classes; 2) CSC: Class-specific context where context vectors are independent to each class; 3) HC_CSC: Hand-crafted prompt "A RS image of a" combined with class-specific context; and 4) UC_CSC: Unified context combine with class-specific context. For the sake of fairness, the total context length of each prompt is 16, and the class-specific context length for the latter two designs is 8.

On the one hand, under the premise of using the same local prompt, we can observe that the global prompt in the form of CSC shows the lowest performance on both datasets, while requiring the most parameters to learn. When the global prompt adopts HC_CSC or UC_CSC, it can bring certain performance improvements compared to CSC. Surprisingly, when using UC for a global prompt, the highest scores are achieved on all metrics in both datasets, significantly surpassing the other three design forms. We speculate that this is because the objective function of the global branch is an image-level classification task, and UC does not require more data for training, resulting in stronger generalization. Therefore, the UC design form is more suitable as the global prompt.

On the other hand, under the premise of using the same global prompt, it can be seen that the performance of the local prompt using UC is the worst. This is because the local prompt needs to learn different features of each category in the first training stage and then guide the learning of visual features related to each category in the second training stage. However, UC cannot fully model fine-grained discrepancies between classes. When using CSC for local prompt, it achieves significant improvement compared to UC, as different features of classes can be described by class-specific contexts. When using HC_CSC for local prompt, it brings further performance gains compared to CSC, because CSC cannot learn common features between classes. The best effect is achieved when using UC_CSC for local prompt, which enables autonomous and flexible modeling of commonality and discrepancy between all categories. Thus, we choose the combination of *UC as the global prompt and UC_CSC as the local prompt* in our implementation.

TABLE VIII

COMPARISON OF DIFFERENT CONTEXT LENGTH COMBINATIONS ON THE POTSDAM AND VAIHINGEN DATASETS. THE VALUES SHOWN ARE (M1, M2), I.E., LENGTHS OF UNIFIED CONTEXT AND CLASS-SPECIFIC CONTEXT. THE BEST RESULTS ARE BOLDED FOR HIGHLIGHTING

Context length	Potsdam			Vaihingen		
	mF1(%)	OA(%)	mIoU(%)	mF1(%)	OA(%)	mIoU(%)
(4,12)	93.72	93.30	88.18	91.64	91.73	84.43
(6,10)	93.76	93.31	88.25	91.75	91.84	84.57
(8,8)	**93.80**	93.35	**88.32**	91.82	**91.98**	**84.65**
(10,6)	93.78	**93.36**	88.28	**91.85**	91.86	84.61
(12,4)	93.78	93.35	88.30	91.80	91.88	84.52

TABLE IX

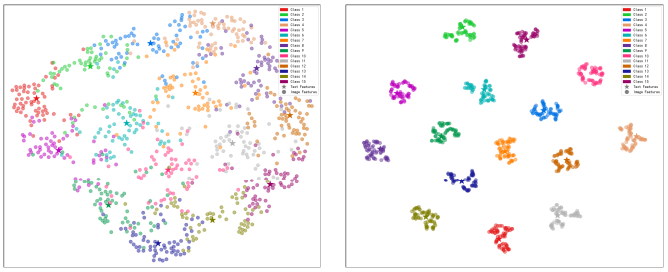
MODEL COMPLEXITY ANALYSIS PERFORMED WITH 256 × 256 IMAGE ON A SINGLE NVIDIA GEFORCE RTX 4090 GPU

Model	Backbone	FLOPs(G)	Parameter(M)	Memory(MB)	Time(s)
AlignSeg [58]	ResNet-101	32.00	334.29	305	0.031
CCNet [54]	ResNet-101	79.71	354.28	357	0.033
EMANet [52]	ResNet-101	57.86	53.77	933	0.009
FarSeg [62]	ResNet-101	11.05	192.07	147	0.007
UperNet [57]	ResNet-101	75.52	522.70	525	0.023
RSSFormer [68]	ResNet-101	71.06	57.33	469	0.011
ABCNet [19]	ResNet-18	7.81	13.67	838	0.097
TransUNet [79]	R50-ViT-B	64.55	105.32	1784	0.077
UNetformer [78]	ResNet-18	5.87	11.69	842	0.084
FTUNetformer [78]	Swin-B	50.84	96.14	2540	0.085
CMTFNet [67]	ResNet-50	17.14	30.07	1496	0.089
SSNet [81]	ViT-B	45.22	85.13	1824	0.088
RS3Mamba [80]	R18-MambaT	28.25	43.32	1824	0.072
Ours(CLIP2RS)	CLIP-B	48.28	87.63	2040	0.083

4) Length of Learnable Tokens: We conduct experiments in Table VIII to explore the influence of different context length combinations. The lengths of unified context and class-specific context tokens are denoted by M1 and M2, respectively. From the results, we can see that the performance is the best when $M1 = 8$ and $M2 = 8$, achieving the highest score on multiple metrics in both Potsdam and Vaihingen datasets. Overall, the performance of our method is not very sensitive to the lengths of context vectors.

5) Model Complexity Analysis: Table IX provides a comparison of the model complexity of our proposed method against several recent approaches, including floating-point operations (FLOPs), model parameters, memory consumption, and the average inference time per image. Although our method employs a vision-language dual-modality pretrained foundation model, the key metrics remain within acceptable ranges while simultaneously achieving promising performance. Although CLIP2RS is not the most lightweight model, its additional computational overhead primarily stems from the dual-modality vision-language backbone and the dual-granularity alignment framework. These architectural designs substantially enhance semantic representation and cross-domain generalization, resulting in superior OA and improved robustness to class imbalance. Therefore, the moderate increase in model complexity is well justified by the significant gains in performance.

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(a) Stage1 (b) Stage2

Fig. 7. t-SNE visualization of image features and text features on the iSAID dataset. Fifteen foreground categories are represented by different colors, and circles and pentagrams indicate the image and text features corresponding to each category. (a) and (b) Data distributions after the first and second training stage.

TABLE X
ANALYSIS OF MIOU(%) RESULTS WITH DIFFERENT LOSS WEIGHT COMBINATIONS ON THREE DATASETS

α(global weight)	β(auxiliary weight)	γ(local weight)	iSAID	Potsdam	Vaihingen
0.1	0.4	0.5	66.3	87.6	83.5
0.2	0.3	0.5	67.6	88.1	84.3
0.3	0.3	0.4	**68.0**	**88.3**	**84.6**
0.5	0.1	0.4	67.1	87.8	83.8
0.5	0.4	0.1	65.9	87.3	82.7
0.6	0.3	0.1	65.5	86.8	82.6

6) t-SNE Visualization Analysis: To better understand the proposed two-stage training strategy and validate its effectiveness, we performed t-SNE visualization analysis of image and text features on the iSAID dataset. The results are shown in Fig. 7, where fifteen foreground categories are represented by different colors, and circles and pentagrams represent the image and text features corresponding to each category. Fig. 7(a) and (b) displays the feature distributions after the first training stage and the second stage, respectively. From the visual results, we can see that after the first training stage, the text features form a scattered distribution based on the image features, generating RS text descriptions with unique semantics for each category. During the fine-tuning process in the second stage, the model further optimizes the alignment between the image and text features. After fine-tuning, the image features of the same category gradually cluster around their corresponding text features, while image features of different categories are effectively separated. This adjustment in feature distribution enhances the discriminative ability between categories while preserving shared characteristics within each category, fully demonstrating that the text features learned in the first stage are effective and suitable for fine-tuning in the second stage.

7) Hyperparameter Analysis of Loss Weights: To obtain the optimal combination of the loss weight coefficients α (global loss weight), β (auxiliary loss weight), and γ (local loss weight), we performed a grid search to evaluate the mIoU results of the model with various hyperparameter combinations across three datasets, as presented in Table X. The analysis indicates that the distribution of weights in the loss function significantly impacts model performance. Specifically, when the hyperparameter combination is [0.3, 0.3, 0.4], the model performs best. However, in the combinations where one of α, β, or γ was set to the lower value of 0.1, the model's performance decreased to varying degrees. This suggests that, for RS semantic segmentation tasks, the local loss weight γ is critical due to the dense prediction nature of the task. Meanwhile, the global loss weight α is essential for capturing global contextual information and alleviating sample imbalance issues. Therefore, a well-balanced allocation of α, β, and γ helps optimize the collaboration between global and local features, further improving the overall model performance.

V. CONCLUSION

In this study, we introduce CLIP2RS, which utilizes the VLM pretrained on natural images for the semantic segmentation of RS images. To bridge the substantial domain gap between natural and RS imagery, CLIP2RS employs a two-phase training approach. The first phase significantly enhances the semantic descriptive capabilities of category labels by generating text prompts appropriate for each RS image category. During the second phase, the image encoder is refined using linguistic prior knowledge embedded within a pixel-text score map. In addition, we introduce a dual-granularity alignment framework that aligns modalities at both pixel and image levels to address class imbalance issues effectively. Finally, we investigate a new prompt design that simultaneously captures both general and category-specific features, maximizing the potential of CLIP textual descriptions. CLIP2RS demonstrates state-of-the-art performance across three extensively used benchmark datasets, underscoring the method's superiority.

CLIP2RS showcases the ability of VLM to boost the accuracy of image segmentation in RS applications. The approach we introduced utilizes static local and global prompts, which might not effectively capture the fine differences among categories. Going forward, we plan to investigate more adaptive methodologies, like dynamic prompting, to enhance the model's performance in intricate RS contexts.

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